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PEC AIML LAB ASSIGNMENT 2

Country Chatbot Using Prolog

Assignment: Rule-Based AI Chatbot

Aim

To design and implement a simple rule-based Country Chatbot using Prolog that can provide information about countries based on user queries.

Objective

The objectives of this experiment are:

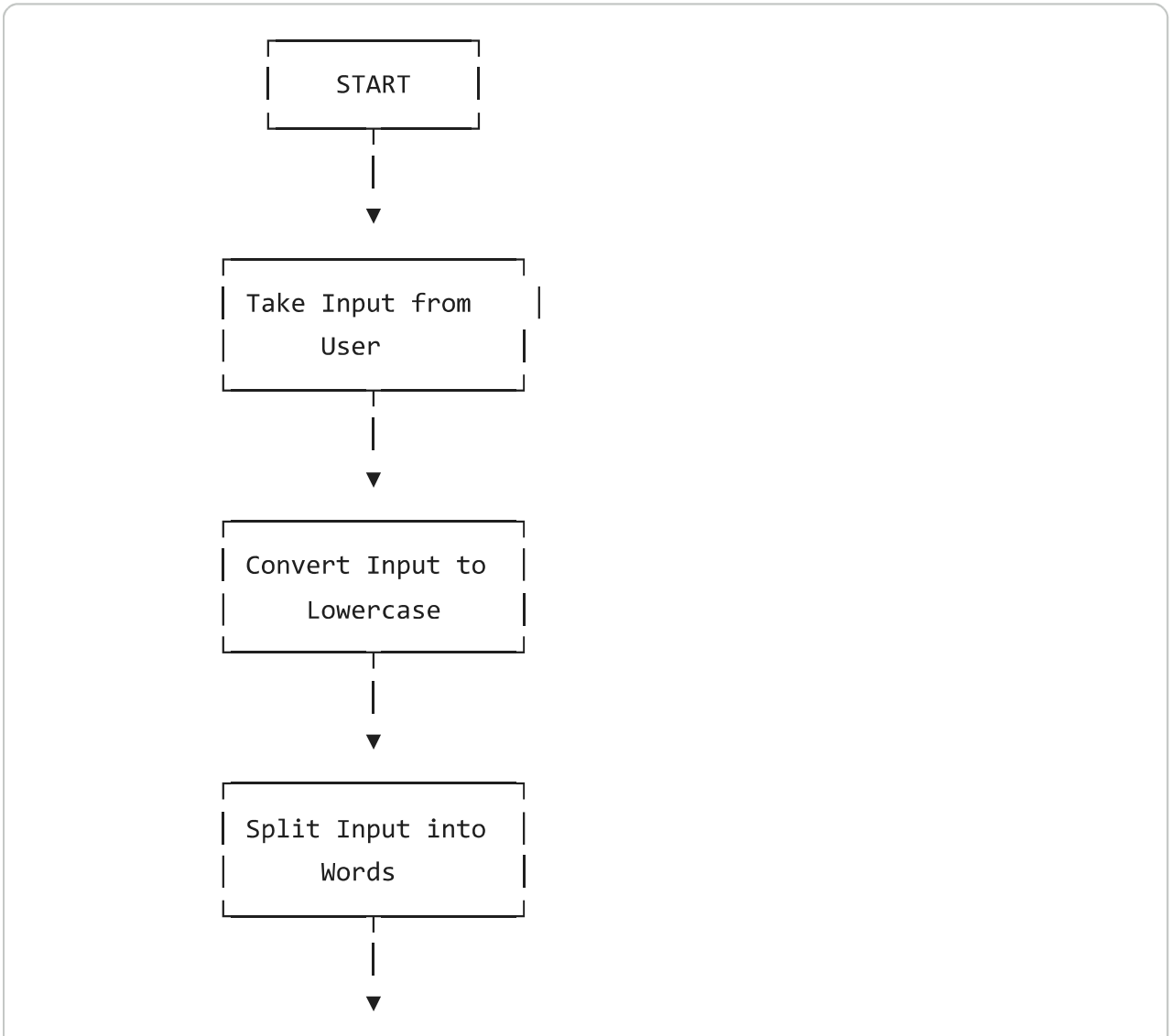
1. To understand the basics of Prolog programming.
2. To implement a simple rule-based chatbot.
3. To use predicates and rules for decision making.
4. To use the member/2 predicate for keyword matching.
5. To provide information about five countries.
6. To answer questions about capital, currency, and continent.
7. To understand user input processing in Prolog.
8. To implement a chatbot that continues until the user enters "quit".

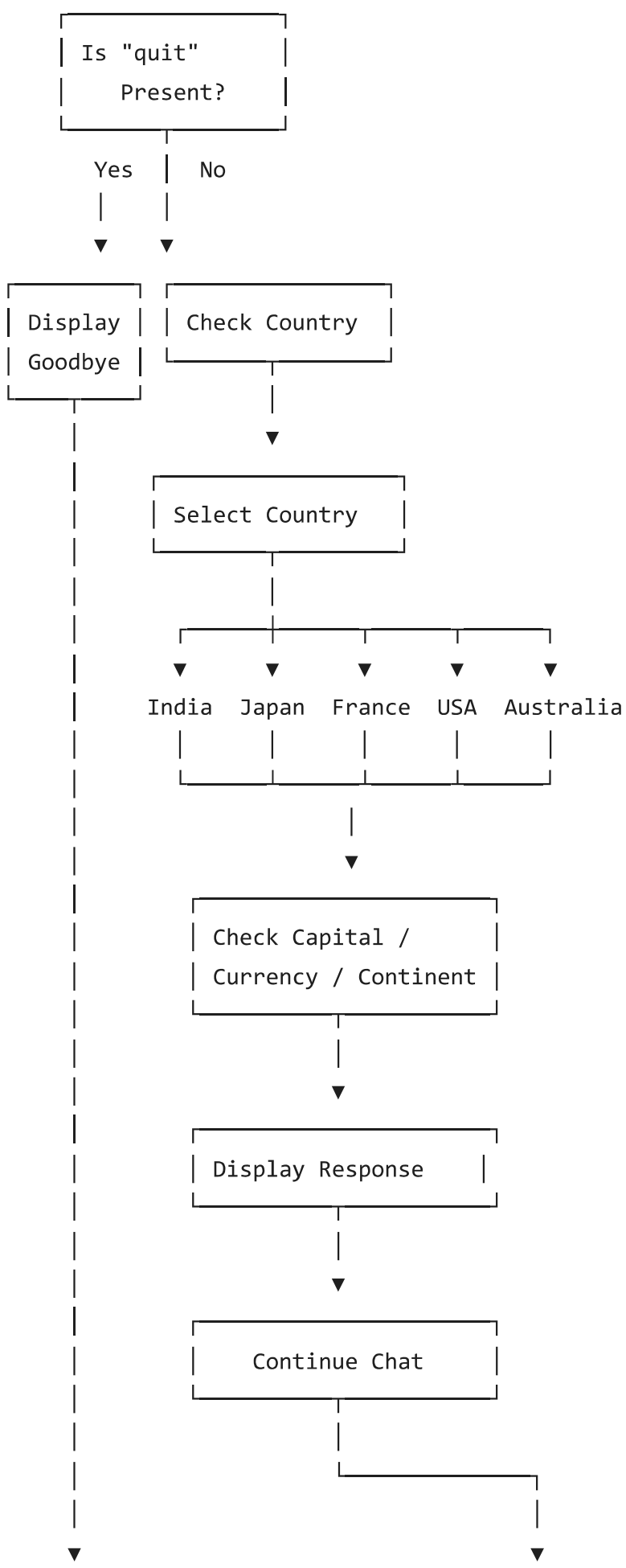
Algorithm

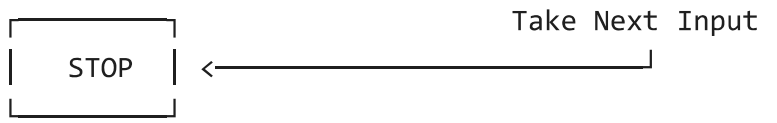
1. Start the program.

2. Display the Country Chatbot welcome message.
3. Take input from the user.
4. Convert the input into lowercase.
5. Split the input into individual words.
6. Check whether the input contains "quit".
7. If "quit" is present, display "Goodbye" and stop.
8. Otherwise, search for the country name.
9. Check whether the query contains "capital", "currency", or "continent".
10. Display the appropriate information.
11. If no matching country is found, display the default response.
12. Repeat the process for the next query.
13. Stop when the user enters "quit".

Flowchart




Is "quit" Present?
Yes
No
Display Goodbye
Check Country
Select Country
India
Japan
France
USA
Australia
Check Capital / Currency / Continent
Display Response
Continue Chat



Result Analysis

The Country Chatbot was successfully implemented using Prolog.

The chatbot was tested with different queries related to five countries: India, Japan, France, USA, and Australia.

The chatbot successfully identified keywords and generated appropriate responses.

Test Results

Query	Expected Response	Result
Capital of India	New Delhi	Successful
Currency of Japan	Yen	Successful
Continent of France	Europe	Successful
Capital of USA	Washington DC	Successful
Currency of Australia	Australian Dollar	Successful
Unknown country	Default response	Successful

The chatbot correctly processes the user input and provides the required information based on the predefined Prolog rules.

PROGRAM CODE

```
chat :-
    writeln('===== COUNTRY CHATBOT ====='),
    writeln('Ask about India, Japan, France, USA, or Australia.'),
    writeln('Ask about capital, currency, or continent.'),
    writeln('Type quit to exit.'),
    nl,

    repeat,
    write('You: '),
    read_line_to_string(user_input, Input),
    string_lower(Input, Lower),
    split_string(Lower, " ", " .?!", Words),
```

```
    ( member("quit", Words)
      ->
      writeln('Bot: Goodbye!'),
      !
      ;
      respond(Words),
      fail
    ).

% India
respond(Words) :-
    member("india", Words),
    ( member("capital", Words) ->
      writeln('Bot: The capital of India is New Delhi.')
    ; member("currency", Words) ->
      writeln('Bot: The currency of India is the Rupee.')
    ; member("continent", Words) ->
      writeln('Bot: India is in Asia.')
    ; writeln('Bot: India is a country in South Asia.')
    ), !.

% Japan
respond(Words) :-
    member("japan", Words),
    ( member("capital", Words) ->
      writeln('Bot: The capital of Japan is Tokyo.')
    ; member("currency", Words) ->
      writeln('Bot: The currency of Japan is the Yen.')
    ; member("continent", Words) ->
      writeln('Bot: Japan is in Asia.')
    ; writeln('Bot: Japan is an island country in East Asia.')
    ), !.

% France
respond(Words) :-
    member("france", Words),
    ( member("capital", Words) ->
      writeln('Bot: The capital of France is Paris.')
    ; member("currency", Words) ->
      writeln('Bot: The currency of France is the Euro.')
    ; member("continent", Words) ->
      writeln('Bot: France is in Europe.')
    ; writeln('Bot: France is a country in Europe.')
    ), !.

% USA
respond(Words) :-
```

```
    member("usa", Words),
    ( member("capital", Words) ->
      writeln('Bot: The capital of USA is Washington DC.')
    ; member("currency", Words) ->
      writeln('Bot: The currency of USA is the US Dollar.')
    ; member("continent", Words) ->
      writeln('Bot: USA is in North America.')
    ; writeln('Bot: USA is a country in North America.')
    ), !.

% Australia
respond(Words) :-
    member("australia", Words),
    ( member("capital", Words) ->
      writeln('Bot: The capital of Australia is Canberra.')
    ; member("currency", Words) ->
      writeln('Bot: The currency of Australia is the Australian Dollar.')
    ; member("continent", Words) ->
      writeln('Bot: Australia is in Oceania.')
    ; writeln('Bot: Australia is a country and continent.')
    ), !.

% Default response
respond(_) :-
    writeln('Bot: Sorry, I do not have information about that country.').

% Queries:
% ?- chat.
% ?- respond(["india"]).
% ?- respond(["india", "capital"]).
% ?- respond(["india", "currency"]).
% ?- respond(["japan", "capital"]).
% ?- respond(["japan", "currency"]).
% ?- respond(["france", "continent"]).
% ?- respond(["usa", "capital"]).
% ?- respond(["australia", "currency"]).
```

Conclusion

The Country Chatbot was successfully implemented using Prolog.

The chatbot uses rules, predicates, keyword matching, conditional statements, repeat, and the cut operator.

It can provide basic information about five countries: India, Japan, France, USA, and Australia.

The chatbot successfully answers questions related to capital, currency, and continent and provides a default response for unknown queries.

Thus, the experiment demonstrates the use of Prolog for developing a simple rule-based Artificial Intelligence chatbot.